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se Code: CSE-253 Course Title: Digital Logic Design

al Examination No: _____ Date: _____ Signature of Course Teacher:

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1. a) $A(B+C) = AB+AC$
2. b) two ANDs ORed together
3. c) $A+B = B+A$
4. d) all of them
5. b) $A(BC) = (AB)C$

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Answers to Q no. 6

Given SOP expression,

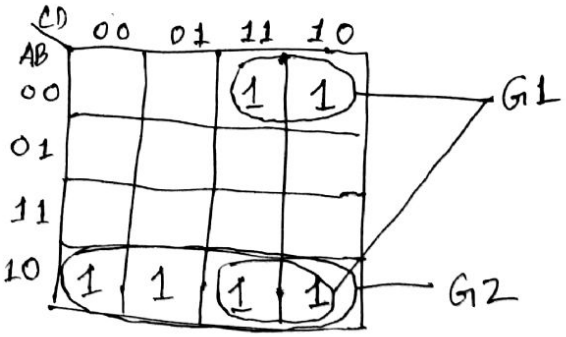
$\bar{B}C + A\bar{B} + A\bar{B}C + A\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}\bar{D}$

0010 1000 1010 1010 1011 1000
0011 1001 1011
1010 ~~1010~~
1011 ~~1011~~

The standard SOP expression is

$\bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + A\bar{B}C\bar{D} + A\bar{B}CD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$

The expression is mapped on a K-Map as follows:



5

The required minimum form from K-Map,

$\bar{B}C + A\bar{B}$

Answers to Q no. 7

Given,

$X = (\bar{A}+B+\bar{C})(A+\bar{B}+C)(\bar{A}+B+C)(A+B+C)$

The truth table is as follows:

2.5

A	B	C	X	~~Product~~ term Sum
0	0	0	0	$(A+B+C)$
0	0	1	1	
0	1	0	0	$(\bar{A}+\bar{B}+\bar{C})(A+\bar{B}+C)$
0	1	1	1	
1	0	0	0	$(\bar{A}+B+C)$
1	0	1	0	$(\bar{A}+B+\bar{C})$
1	1	0	1	
1	1	1	1	

Answer to Q no. 8(a)

Given,

$\bar{A}BC + A\bar{B}\bar{C} + \bar{A}\bar{B}\bar{C} + A\bar{B}C + ABC$

$= \bar{A}BC + A\bar{B}(\bar{C}+C) + \bar{A}\bar{B}\bar{C}$

$= \bar{A}BC + A\bar{B} + \bar{A}\bar{B}\bar{C}$; [$\because \bar{C}+C = 1$ and $A\bar{B}\cdot 1 = A\bar{B}$]

$= \bar{A}BC + \bar{B}(A+\bar{A}\bar{C})$

$= \bar{A}BC + \bar{B}(A+\bar{C})$; [$\because A+\bar{A}\bar{C} = A+\bar{C}$]

$= \bar{A}BC + A\bar{B} + \bar{B}\bar{C}$; [Distributive Law]

$= \bar{A}BC + A\bar{B} + \bar{B}\bar{C}$

$=$

Answer to Q no. 8(b)

Given,

$A\bar{B} + A\bar{B}\bar{C} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$

$= A\bar{B}(1+\bar{C}) + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$

$= A\bar{B} + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}E$; [$\because 1+\bar{C} = 1$ and $A\bar{B}\cdot 1 =$

$= A\bar{B}(1+\bar{C}\bar{D}) + A\bar{B}\bar{C}\bar{D}E$

$= A\bar{B} + A\bar{B}\bar{C}\bar{D}E$

$= A\bar{B}(1+\bar{C}\bar{D}E)$

$= A\bar{B}$

Answer to Q no. 9(a)

Given,

$\overline{A+B\bar{C}} + D\overline{(E+\bar{F})}$

$= (\overline{\overline{A+B\bar{C}}}) \cdot \overline{D(\overline{E+\bar{F}})}$; [Using De Morgan's Law]

$= (A+B\bar{C})(\bar{D} + (\overline{\overline{E+\bar{F}}}))$; [Using De Morgan's Law]

$= (A+B\bar{C})(\bar{D}+E+\bar{F})$

Answer to Q no. 9(b)

Given,

$\overline{ABC} + D + F$

$= (\overline{\overline{ABC}}) \cdot \bar{D} \cdot \bar{F}$; [Using De Morgan's Law]

$= ABC\bar{D}\bar{F}$

Answer to Q no. 10(a)

Given,

$(A+B)(B+C)$

$= (A+B+C\bar{C}^{\bullet})(A\bar{A}^{\bullet}+B+C)$

$= (A+B+C)(A+B+\bar{C}^{\bullet})(A+B+C)(\bar{A}^{\bullet}+B+C)$

$= (A+B+C)(A+B+\bar{C}^{\bullet})(\bar{A}^{\bullet}+B+C)$

Answer to Q no. 10(b)

Given,

$\bar{A}BC + \bar{A}B\bar{C} + A\bar{B}C + ABC$

$011 \quad 010 \quad 101 \quad 111$

The missing terms among the $2^3 = 8$ c
are 000, 001, 100, 110.

∴ The required SPOS expression,

$(A+B+C)(A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$